More Unlabeled Data Does Matter: A Full-Cycle Framework for Semi-Supervised Semantic Segmentation of Remote Sensing Images

Li Zhang, Zhenshan Tan, Wen Zhang, Zhiwei Li, Guo Zhang, Zefeng Chen, and Zhijiang Li

***Abstract*— Semi-supervised semantic segmentation has gained considerable attention due to its ability to leverage large amounts of unlabeled data to enhance model generalization. Although increasing the amount of unlabeled data ideally enhances performance, reality contradicts this notion. Research indicates that the reasons for performance degradation span the entire semi-supervised learning process. These include the inconsistent distribution of unlabeled and labeled data at the data level, the susceptibility to interference from unknown categories at the model level, and the semantic drift problem caused by the training strategy. This article refers to these challenges as the "full-cycle" challenge. To address these issues, we propose a full-cycle framework. At the data level, we develop a consistency-based method, beginning with unlabeled data screening, to prioritize unlabeled data that closely aligns with the labeled data distribution in training. At the model level, we introduce a consistency-based semi-supervised semantic segmentation model capable of identifying unknown category regions and distancing their features from known category features, reducing the interference of unknown classes. At the training strategy level, we adopt a progressive approach that gradually incorporates data from simpler to more complex cases. This strategy improves the model's adaptability to potential data changes, ensures stability, and improves overall generalization performance. Extensive experiments demonstrate that our method significantly outperforms existing state-of-the-art approaches, validating its effectiveness in addressing the full-cycle challenge.**

***Index Terms*— Data distribution, remote sensing images, semantic drift, semi-supervised semantic segmentation, unknown categories.**

I. Introduction

SEMANTIC segmentation of remote sensing images is essential for applications such as land cover classification [1], [2], [3], [4]; however, the annotation process is costly and time-consuming, leading to a scarcity of labeled data [5]. To address this limitation, recent research has increasingly focused on leveraging unlabeled data through semi-supervised [6], [7], weakly supervised [8], [9], [10], and unsupervised [11] semantic segmentation methods. Since existing labeled data provide high-quality supervision signals that enhance model generalization, they are a critical foundation for effectively incorporating unlabeled data. Building on this insight, this study investigates the performance of semi-supervised semantic segmentation.

Most of the existing studies optimize semi-supervised learning performance by adjusting the ratio of labeled to unlabeled data while keeping the total data volume fixed [12], [13]. Although this approach provides valuable insights into model training under constrained data resources, it does not fully align with real-world scenarios, where labeled data is often scarce, while vast amounts of unlabeled data are readily available. To better reveal practical applications, we adopt a more realistic setting by fixing the amount of labeled data and progressively increasing the amount of unlabeled data. As shown in Fig. 1(a), we evaluate model performance with unlabeled data amounts set to 1, 3, 7, and 15 times the labeled data (the horizontal axis in the figure represents the proportion of labeled data to the total data). The figure shows that model performance decreases significantly when the unlabeled data is 7 or 15 times larger than the labeled data. This phenomenon reflects a practical limitation in real-world applications such as land cover classification, post-disaster assessment, and urban change detection, where models are expected to handle vast remote sensing archives with minimal manual annotation [14], [15]. In such scenarios, the inability to effectively utilize abundant unlabeled data not only limits model scalability but also risks producing misleading results, especially in safety-critical tasks. This finding suggests that current models struggle to achieve significant performance gains solely by adding unlabeled data, which may limit their scalability in practical applications. Further investigation reveals that per-

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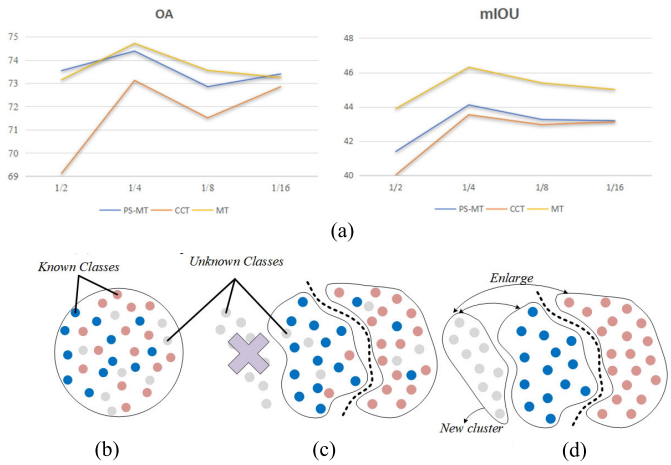


Fig. 1. (a) Change in accuracy across different models as the amount of unlabeled data increases, with the horizontal axis representing the proportion of labeled data. (b) Feature composition of unlabeled data, where blue and orange represent features of known categories, while gray represents features of unknown categories. (c) Previous method's treatment of unknown categories. The purple cross indicates that the previous method directly filters out the unknown category. (d) Our method's treatment of unknown categories. Identify unknown categories and distinguish them from known category features.

formance degradation stems from three key aspects: data [16], [17], model [18], [19], and training strategy [20], [21]. These three factors permeate the semi-supervised learning process, which is called the "full-cycle" challenge in this article.

First, at the data level, the semi-supervised semantic segmentation tasks assume that labeled and unlabeled data follow the same distribution. However, in real-world applications, this assumption often fails. Data with a substantial distribution divergence from the labeled data can severely undermine the model's generalization ability, leading to suboptimal performance in real-world applications. Thus, identifying unlabeled data that closely mimics the distribution of labeled data is imperative for enhancing the model's performance in complex remote sensing scenarios. Existing methods improve the model's adaptability across different scenes and enhance its generalization through confidence screening mechanisms [22], [23] and data augmentation techniques [24]. However, most of these approaches overlook the challenge of systematically improving the quality of unlabeled data at the source in large-scale unlabeled data scenarios. Second, at the model level, current semi-supervised semantic segmentation models struggle to effectively handle the interference of unknown categories. This is because pseudo-label mechanisms typically assume that unlabeled data shares the same category set as labeled data. However, due to the geographical diversity in remote sensing images, labeled data often covers only a subset of categories, while unlabeled data may contain additional, unseen categories. As a result, unknown categories in the unlabeled data are prone to being misclassified as known categories, reducing segmentation accuracy [as shown in Fig. 1(b)]. Previous studies [25], [26] aim to mitigate this issue by identifying and excluding unknown categories from consistency training. However, due to the high dimensionality and complexity of remote sensing images, existing methods often fail to detect all unknown categories, leading to misclassifications that interfere with the learning of known categories and further degrade overall performance [as shown in Fig. 1(c)]. Finally, at the training strategy level, inappropriate handling of large-scale unlabeled data during training can exacerbate semantic drift, a phenomenon where the model's feature space gradually misaligns due to the noisy or inconsistent pseudo-labels generated from unlabeled data. This issue is particularly pronounced in remote sensing images, where high intra-class variability increases the risk of feature misalignment. Without explicit supervision, the model may inadvertently adapt its feature distribution to accommodate variations in the pseudo-labeled data, further amplifying semantic drift. To address this, progressively introducing high-confidence pseudo-labels is typically employed [12] to support steady performance improvement and prevent further pseudo-label errors. However, it may rely excessively on simpler samples in the early stages, potentially causing the model to converge prematurely to a local optimum and limiting its generalization to more complex samples. These issues mentioned above do not occur in isolation throughout the training process but instead influence one another. As the model iterates, these problems accumulate at various stages, ultimately affecting its overall performance. Focusing on optimizing a single component alone cannot fully address the complexity introduced by unlabeled data.

Motivated by the challenges posed by large-scale unlabeled data, we propose a comprehensive full-cycle semi-supervised semantic segmentation framework. Unlike previous works that target isolated components of the pipeline, our method addresses the interdependent challenges across the entire semi-supervised learning process in a unified, full-cycle framework. This includes data-level screening to ensure input quality, model-level separation of unknown categories to protect learning stability, and strategy-level curriculum design to mitigate semantic drift.

First, we begin by screening unlabeled data by designing a consistency measurement module (CMM) to assess the distribution similarity between unlabeled and labeled data. Using this module, we calculate a score for each unlabeled image, rank, and select suitable data based on these scores. This meticulous selection of unlabeled data can serve as a linchpin in optimizing the semi-supervised learning process, enabling the model to better adapt to the multifaceted nature of remote sensing data and achieve more accurate and robust semantic segmentation results, and reduce the influence of unknown class interference and semantic drift to a certain extent. Second, we develop a novel consistency-based semi-supervised segmentation network that specifically tackles the degradation of model performance caused by unknown categories. This architecture features a dedicated classifier that identifies pixels belonging to unknown categories, enabling us to isolate these regions. We then increase the separation between known and unknown category features in the feature space and apply consistency learning solely to known category regions. This approach minimizes the impact of unknown categories in unlabeled data on the model's learning process. Finally, we propose a batch training strategy for the semantic drift problem. In the early stages of training, we prioritize unlabeled data that closely aligns with the distribution of

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labeled data. This approach ensures the model learns within a clear and unambiguous semantic context, facilitating the establishment of accurate decision boundaries. As training progresses, we gradually introduce more complex unlabeled data, which are relatively inconsistent with the distribution of labeled data, to enhance the model's robustness. This progressive training strategy, starting with easier examples before transitioning to more challenging ones, effectively mitigates the risk of semantic drift when the model encounters complex unlabeled data. It maintains model stability while improving adaptability to potential data changes, ultimately improving generalization performance. Unlike previous methods that fix the total amount of data, we introduce a more flexible setting by fixing the amount of labeled data, which is better suited for complex real-world scenarios. Specifically, the main contributions of this article are summarized as follows.

1) A full-cycle semi-supervised semantic segmentation framework is proposed, systematically addressing the challenges posed by large-scale unlabeled data through data selection, model design, and training strategies.
2) A CMM is designed to optimize the selection and utilization of unlabeled data by assessing the distribution similarity between unlabeled and labeled data.
3) A dedicated classifier is introduced to identify unknown categories, expanding the distinction between known and unknown categories in feature space and reducing the interference of unknown categories on the model.
4) A progressive training strategy is proposed, prioritizing unlabeled data that closely aligns with the labeled data distribution in the early stages, effectively mitigating semantic drift and enhancing model robustness and generalization.

II. Related Work

A. Screening of Training Data

The performance of semi-supervised semantic segmentation is strongly influenced by data quality. In practice, the quality of unlabeled data varies greatly, and low-quality samples would severely impact the model's segmentation accuracy. Recent strategies for improving the quality of unlabeled data in semi-supervised semantic segmentation can be broadly categorized into confidence screening mechanisms and data enhancement techniques [15], [27]. Sohn et al. [28] proposed the FixMatch framework, which integrates consistency regularization and a confidence threshold to utilize only high-confidence pseudo-labels during training. This approach effectively mitigates noise from unreliable pseudo-labels and enhances segmentation performance. Chen et al. [29] introduced a cross-pseudo-supervision approach with a dual-model architecture to enable the models to learn complementary features, effectively mitigating the impact of noisy labels. Additionally, CutOut [30], CutMix [31], and ClassMix [32] generate new images by combining two images in distinct ways, serving as classic data augmentation techniques widely applied in semi-supervised learning to improve the generalization of semi-supervised semantic segmentation models. Xin et al. [12] proposed a semi-supervised semantic segmentation framework that integrates consistency regularization and contrastive learning (CL). By assigning higher bias weights to more challenging negative samples, their approach increases the complexity of feature learning and enhances the discriminative capability of the final feature representation space. Similarly, Qi et al. [33] employed a selective self-training strategy by generating and selecting high-quality pseudo-labeled samples, ensuring consistency during model training to achieve improved segmentation performance. These methods constrain unlabeled data at the feature level; however, they may become ineffective when the quality of the samples is excessively poor.

Additionally, several studies have examined labeled data screening in semi-supervised semantic segmentation. Yuan et al. [34] proposed ActiveMatch, which integrates representation learning with active sample selection to prioritize labeling data that is highly informative and minimally noisy, ensuring that only the most valuable data contributes to the learning process. Gao et al. [35] introduced an active learning screening strategy in which samples with low consistency between their original and perturbed predictions indicate limited value to the semi-supervised learning model, thus requiring manual labeling. Similarly, Wang et al. [36] applied unsupervised techniques to selectively label only the most valuable samples, improving semi-supervised semantic segmentation performance by minimizing uninformative or noisy data.

While the aforementioned methods improve semi-supervised semantic segmentation performance through unlabeled data enhancement, high-confidence feature filtering, and labeled data screening, few studies address the quality of the original unlabeled data itself, particularly from the perspective of distributional gaps between labeled and unlabeled data. This gap can significantly impact model performance, yet it remains underexplored in the existing literature.

B. Semi-Supervised Semantic Segmentation Model

Current mainstream methods for semi-supervised semantic segmentation can be broadly categorized into two main approaches: self-training [19], [37], [38], [39] and consistency-based [40], [41], [42], [43] methods. To mitigate the impact of unknown categories on semantic segmentation networks, Mundt et al. [25] analyzed the prediction uncertainty of different model types and methods for detecting unknown categories based on extreme-value theory (EVT). Huang et al. [44] applied prototype kernel learning combined with open-set foreground perception for generalized small-target semantic segmentation, leveraging kernel-based methods to maintain consistent representations between old and novel classes. Xie et al. [45] proposed a prototype network that enhances model discriminative capability by incorporating a contrastive clustering module and designed a prototype-guided open set module (POSM) to distinguish between known and unknown categories. Tang et al. [46] introduced a holistic prototype attention network (HPAN) that transfers knowledge from known to unknown classes by employing a prototype graph

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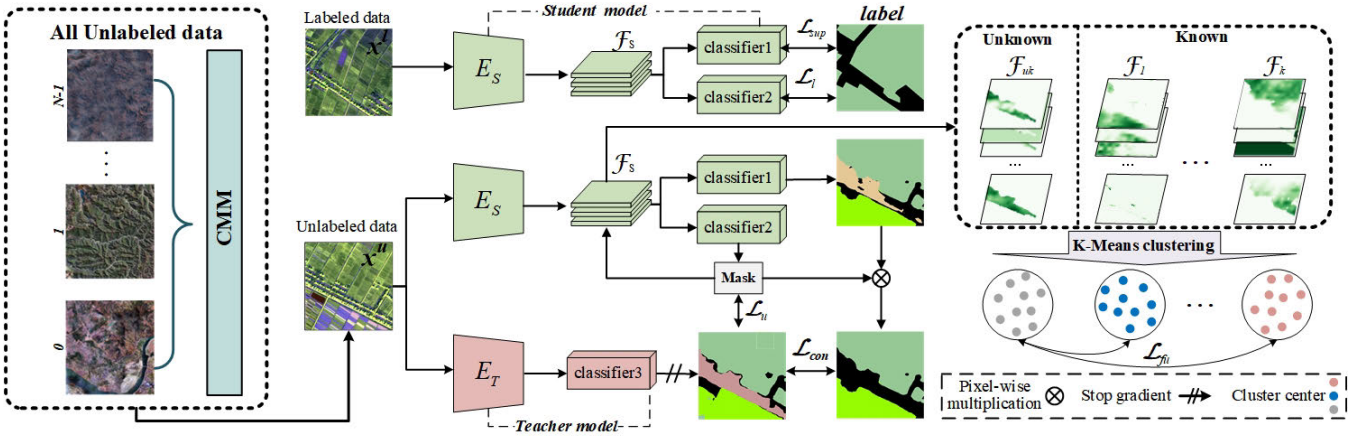


Fig. 2. Architecture of our proposed network. On the left, the CMM represents the screening mechanism for unlabeled data. In the middle section, identical colors in the first and second rows indicate shared network parameters. Classifier 1 is the main classifier, while classifier 2 is an auxiliary classifier to locate unknown categories. On the right, F_{uk} denotes features associated with unknown categories, while $F_1, \ldots, F_k$ represent features for each known category. Colored dots indicate the cluster centers of different categories, with the number of dots corresponding to the number of channels in the feature map.

attention module (PGAM) and a bidirectional prototype attention module (BPAM). Zhao et al. [26] proposed novel class discovery in semantic segmentation (NCDSS), which utilizes labeled base data and saliency models to cluster new classes coarsely. Cha et al. [47] developed a class-imbalanced sampling strategy using a visual saliency detector to filter out unknown classes. While these methods have made notable progress in addressing the issue of unknown categories, many current solutions are primarily applied in open-set learning scenarios [48], [49], where the goal is to identify and separate both known and unknown categories. However, in the context of semi-supervised semantic segmentation for remote sensing imagery, which generally operates in a closed-set setting, these challenges have not been extensively explored. Remote sensing data presents unique complexities, such as geographical variability, seasonal changes, and the diversity of environmental features, that make the handling of unknown categories a particularly significant issue in closed-set learning. We aim to adapt and apply ideas from open-set learning to the semi-supervised semantic segmentation of remote sensing imagery, in order to mitigate the interference of unknown categories while maintaining the closed-set nature of the problem.

Another major challenge in semi-supervised semantic segmentation is semantic drift. To address this, Xie et al. [50] introduced a memory-inspired semantic segmentation augmentation network (MiSSNet) that implicitly mitigates category semantic drift by constructing a series of pixel-level correlation matrices. Zhao et al. [51] reduced semantic drift and classifier bias by selecting specific anchor points from the previous model and positive and negative embeddings from the current model, applying CL in this asymmetric latent space. Cermelli et al. [52] proposed reducing semantic drift by modeling latent classes. While these studies make significant contributions, they mainly focus on class-incremental learning. In contrast, addressing semantic drift in semi-supervised semantic segmentation remains a more pressing challenge, especially when there is a lack of sufficient labeled data to prevent the model from drifting toward incorrect class distributions. The current literature often overlooks the impact of semantic drift in semi-supervised settings, highlighting the need for further research on robust training strategies that minimize its effects and ensure model stability during training.

III. Methods

A. Network Architecture

To effectively utilize unlabeled data while mitigating semantic drift, the proposed framework integrates a CMM and a consistency-based semi-supervised semantic segmentation (CB4S) network. As shown in Fig. 2, the CMM first evaluates the distribution similarity between unlabeled and labeled data, selecting a subset of high-quality unlabeled samples. These selected samples, along with labeled data, are then fed into the CB4S network. Within CB4S, the teacher model generates initial predictions, while an auxiliary classifier identifies unknown regions, ensuring consistency learning is applied only to known categories. This separation helps reduce interference from unknown regions, enhancing model robustness. Additionally, a progressive training strategy is adopted to refine decision boundaries and mitigate semantic drift. The specific process is as follows.

Given a labeled dataset $D_L = \{x_i^l, \text{label}_i\}_{i=0}^{M}$, M represents the number of labeled data. Corresponding to a large number of unlabeled datasets $D_U = \{x_i^u\}_{i=0}^{N}$, where N represents the number of unlabeled data and $M << N$. This article addresses full-cycle training for semi-supervised semantic segmentation in remote sensing imagery. As illustrated in Fig. 2, the network begins by screening unlabeled data through a dedicated mechanism

$$D_u = \left\{x_i^u\right\}_{i=0}^{n} = \text{CMM}(D_U), \quad D_u \in D_U, \ n < N. \tag{1}$$

The screened data is then input into the model, supporting efficient and effective training. First, it is fed into the teacher model to obtain its predictions

$$P_T^u = \text{classifier3}\left(E_T\left(x^u\right)\right). \tag{2}$$

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Among them, E_T and classifier3 represent the teacher model's encoder and classifier. P_T^u represents the prediction result of the teacher model. The student model is trained using both labeled and unlabeled data. First, it is necessary to identify the unknown category area

$$\text{mask} = \text{argmax}(\text{classifier2}(E_S(x))), \quad x \in \{D_L, D_u\}. \tag{3}$$

E_S and classifier2 represent the teacher model encoder and auxiliary classifier, respectively. The mask is used to find the position of the unknown category and to separate the unknown category features from the known category features

$$[F_1, \ldots, F_k] = F_S^u * \text{mask}, \quad F_{\text{uk}} = F_S^u * (1 - \text{mask}). \tag{4}$$

Among them, k represents the total number of known categories, $F_S^u = E_T(x^u)$ represents the characteristics of unlabeled data, which is input into the main classifier and combined with the mask to calculate the final prediction P_S^u

$$P_S^u = \text{classifier1}\left(F_S^u\right) * \text{mask}. \tag{5}$$

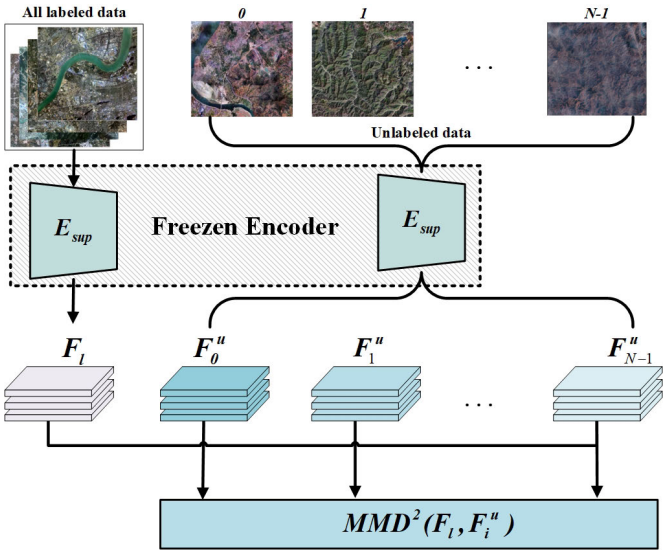


Fig. 3. Architecture of unlabeled data screening based on CMM.

B. Unlabeled Data Screening Based on Distribution Consistency Measurement

In semi-supervised learning, the quality of unlabeled data is crucial, as the model's generalization heavily relies on the representativeness of this data. Ideally, both labeled and unlabeled data should share the same distribution. However, in semi-supervised remote sensing semantic segmentation, bridging the distribution gap between labeled and unlabeled data is an extremely arduous task. This is mainly attributed to a constellation of factors. Geographical spatial heterogeneity plays a pivotal role, as different regions exhibit distinct characteristics. Temporal dynamics also present significant challenges. Seasonal oscillations trigger cyclic changes in vegetation phenology, and the area of the water body varies markedly between the wet and dry seasons. Additionally, the continuous evolution of man-made features, such as the construction of new roads and the expansion of urban areas, creates a pronounced temporal semantic chasm between historical labeled data and contemporary unlabeled data. The class imbalance in the labeled data further compounds the problem. Low-frequency class features, like rare vegetation species or diminutive man-made structures, are inadequately represented in labeled data. Conversely, they might be relatively abundant in unlabeled data, thereby hampering the model's proficiency in discerning these long-tail class features. Within the semi-supervised learning paradigm, the limitations of the pseudo-labeled generation mechanism can magnify distribution discrepancies. During the training process, the model may generate substandard pseudo labels for out-of-distribution samples in the unlabeled data, which can misdirect the feature learning trajectory and precipitate a negative feedback loop.

Inspired by the assumption that unlabeled data and labeled data share the same distribution, we propose a data screening method that leverages feature distribution consistency. This method identifies the most relevant unlabeled samples by assessing their distributional similarity to labeled data and filtering out low-similar data to ensure that the chosen samples enhance the learning process. In this way, the unlabeled data involved in training is guaranteed to meet the above benchmark assumptions to the greatest extent. The specific method is shown in Fig. 3. Initially, we use labeled data to train a baseline semantic segmentation model, denoted as $D_{\text{sup}}(E_{\text{sup}}(.))$. Upon completing the training, we freeze the parameters of feature extractor E_{sup} and use it to obtain the feature representation F_l from the labeled sample. This process enables us to construct a distribution of labeled data. For each unlabeled image x_i^u within the unlabeled dataset D_U, we apply the same feature extractor to derive its feature representation F_i^u

$$F_l = E_{\text{sup}}\left(x^l\right), \quad F_i^u = E_{\text{sup}}\left(x_i^u\right). \tag{6}$$

We employ the maximum mean discrepancy (MMD) [53] metric to compare the labeled data distribution with each unlabeled sample's feature distribution. The consistency of the feature distribution between each unlabeled sample and the labeled dataset is calculated as follows:

$$\begin{aligned} \text{score}[i] &= 1 - \text{MMD}^2\left(D_L, x_i^u\right) \\ &= 1 - \left\| \frac{1}{N_L} \sum_{i=1}^{N_L} \phi(F_l) - \phi\left(F_i^u\right) \right\|_{\mathcal{H}}^2. \end{aligned} \tag{7}$$

Among these, $\phi(.)$ maps the feature vector to the reproducing kernel Hilbert space (RKHS) $\mathcal{H}$. A lower MMD score indicates a greater similarity between the distribution of the unlabeled image and that of the labeled dataset D_L. By calculating the MMD for all unlabeled samples, we rank them according to their similarity to the labeled data and selectively screen those with higher scores for training as required. Despite the inherent disparities present in unlabeled data, maximized efforts have been made to ensure feature distribution alignment with labeled data, thereby partially mitigating interference from unknown categories and alleviating the effects of semantic drift.

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C. Consistency-Based Semi-Supervised Semantic Segmentation Network

Remote sensing images cover vast geographical areas and exhibit high spatial heterogeneity. In situations with limited labeled data, the labeled dataset typically represents only a subset of land cover types, while the unlabeled data may originate from diverse environments (e.g., different geographic regions, seasons, etc.), potentially containing categories not present in the labeled dataset. Interference from unknown categories presents a significant challenge in the semi-supervised semantic segmentation of remote sensing images. It may create confusion in the feature space, leading the model to mistakenly merge features from unknown categories with those from known categories, thereby distorting category representation and decision boundaries.

We design a model capable of effectively extracting unknown category regions to address this issue. To mitigate interference caused by feature similarity between unknown and known categories, we introduce a mandatory constraint that ensures the distance between the features of unknown categories and those of each known category gradually increases. As the training process continues to iterate, this constraint helps reduce the impact of unknown categories, ultimately improving the model's classification performance. The specific network architecture is as follows.

Following the filtering of unlabeled data according to Section III-B, we design a new CB4S network architecture (illustrated in the central section of Fig. 2) comprising a teacher model and a student model. The encoder and decoder structures of the teacher model align with the encoder and main decoder structures of the student model. The teacher model generates pseudo-labels to support the training of the student model, with its parameters updated by those of the student model

$$\theta_t = \alpha\theta_{t-1} + (1-\alpha)\theta_s \tag{8}$$

where θ_t represents the parameters of the teacher model at the time step t, θ_s represents the parameters of the student model, and $\alpha \in [0, 1]$ is a smoothing coefficient.

The core idea of this article for handling unknown categories is to identify and separate unknown categories in the feature space, applying consistency learning exclusively to known category regions to minimize the impact of unknown categories on model performance. The student model's classifier is divided into a main classifier (classifier 2 in Fig. 2) and an auxiliary classifier (classifier 1 in Fig. 2). Unlike the main classifier, which predicts semantic labels among the known classes, the auxiliary classifier is trained to distinguish whether each pixel belongs to the known or unknown category set. During training, we supervise the auxiliary classifier with two types of supervision signals. For labeled data, ground-truth annotations are used to indicate known-category regions. For unlabeled data, we adopt a confidence-based strategy: predictions with confidence lower than a predefined threshold are treated as potential unknown regions, while high-confidence predictions are considered known. This enables the auxiliary classifier to learn a decision boundary between known and unknown regions using both labeled and unlabeled data. Specifically, if the input is a labeled feature, the supervision signal is $y = 1$; if the input is an unlabeled feature, the supervision signal is defined as follows:

$$\mathbf{y} = \begin{cases} 1, & \text{confidence} > T \\ 0, & \text{confidence} < T. \end{cases} \tag{9}$$

The auxiliary classifier is designed to identify the location of unknown categories and to classify the unlabeled data features into known and unknown category features. The features from the encoder are shared between the main and auxiliary classifiers. To ensure that the auxiliary classifier provides reliable identification of unknown regions, we further introduce a separation loss in the feature space. Specifically, we push the features of predicted unknown regions away from the centroids of known class features. This enforces a clear margin between known and unknown representations, reducing the risk of misclassification. The cluster center c_i, $i = [1, K]$ for each known category region is computed using the K-means clustering algorithm [54] and the feature center for the unknown category region c_u. Cosine similarity is then used to assess the similarity between these features. To maximize the separation between unknown category features and the features of each known category, we define the loss function as follows:

$$\mathcal{L}_{\text{fu}} = \frac{1}{K-1}\sum_{i=2}^{K} \max(0, \cos(\mathbf{c}_i, \mathbf{c}_u) - \text{margin}). \tag{10}$$

Finally, as shown in (5), only the features of known category regions are used when applying consistency constraints to the student model.

D. Optimization of Training Strategy

Remote sensing images often exhibit significant intra-class variability, meaning that the same class may display distinct spectral or textural features under different geographical regions, times, or imaging conditions. When unlabeled data is introduced, the model may lack clear supervisory signals, leading to potential misalignment in the feature distribution to accommodate these variations, resulting in semantic shift. This shift occurs as the model gradually deviates from the correct class discrimination standards, which negatively affects segmentation accuracy.

The semantic drift becomes particularly pronounced in scenarios with large intra-class variability, where the same class of objects may show considerable differences in spectral reflection values across different times or regions. These variations can confuse the model, causing it to misclassify certain instances of a class as another, further degrading the segmentation performance. This exacerbates the challenge of maintaining stable classification boundaries, ultimately diminishing the model's generalization ability and segmentation accuracy in remote sensing applications.

To alleviate the semantic drift problem in semi-supervised semantic segmentation, we propose an optimized training strategy using a progressive, batch-based data input approach. Following the unlabeled data ranking in Section III-B, the initial training phase prioritizes high-scoring unlabeled data,

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which better aligns with labeled data distributions, to enable learning within a simpler semantic environment. Early in training, this focus on high-score data enhances the model's ability to distinguish known categories while reducing potential performance loss from uncertain pseudo-labels or unknown categories. As training advances, lower-scoring, more complex unlabeled data are gradually incorporated, increasing the training complexity over time. The gradual increase in difficulty allows the model to progressively adapt to more ambiguous and challenging samples, thereby improving its generalization ability. The specific process is outlined below

$$D_{\text{current}} = \left\{x_i^u\right\}_{i=m}^{m+\text{increment}}, \quad D_{\text{current}} \in D_u \tag{11}$$

$$m = \text{inital}_{\text{amount}} + \text{increment} * \left(\text{epoch}_{\text{current}} - 1\right). \tag{12}$$

Here, m denotes the initial data index for each epoch, $\text{inital}_{\text{amount}}$ denotes the initial data volume, $\text{increment} = (T - \text{inital}_{\text{amount}})/E$ represents the additional data volume per epoch and we define it dynamically based on the total number of training epochs (E) and the total amount of unlabeled data (T). $\text{epoch}_{\text{current}}$ represents the current epoch.

This progressive training strategy reduces the risk of overfitting in the model's early stages and mitigates semantic drift caused by accumulating erroneous pseudo-labels. When applied to large-scale unlabeled datasets, the model can gradually handle complex real-world data, enhancing generalization across diverse data distributions and significantly improving overall robustness and adaptability.

E. Loss Function

The objective function of the proposed semi-supervised semantic segmentation model integrates multiple loss components to ensure effective learning from both labeled and unlabeled data. It is defined as follows:

$$\mathcal{L} = \mathcal{L}_{\text{sup}} + \mathcal{L}_{lu} + \beta \cdot \mathcal{L}_{\text{con}} + \gamma \cdot \mathcal{L}_{\text{fu}}. \tag{13}$$

Here, $\mathcal{L}_{\text{sup}}$ denotes the supervision loss, primarily used to supervise the student model with labeled data. $\mathcal{L}_{lu}$ represents the loss from supervising classifier 2 using labeled and unlabeled data. $\mathcal{L}_{\text{con}}$ refers to the consistency loss function, which ensures stability and consistency in the model's predictions on unlabeled data. $\mathcal{L}_{\text{fu}}$ denotes the feature loss between known and unknown categories. β and γ represent the respective weights of these losses. β and γ represent the respective weights of these losses. In our experiments, both β and γ are empirically set to 1. This configuration follows common practice in semi-supervised learning, where equal weighting is often adopted when there is no prior assumption about the relative importance of individual loss components.

The supervised loss $\mathcal{L}_{\text{sup}}$ is computed using a multiclass cross-entropy function to train the segmentation model on labeled data

$$\mathcal{L}_{\text{sup}} = -\frac{1}{N}\sum_{i=1}^{K}\sum_{c=1}^{K} y_{ic}\log\left(p_{i,c}\right) \tag{14}$$

where K is the number of known classes, y_{ic} is the one-hot ground-truth label for sample i, and $p_{i,c}$ is the predicted probability for class c.

As shown in Fig. 2, the auxiliary classifier (classifier 2) is supervised by both labeled data and unlabeled data, so its loss function is as follows:

$$\mathcal{L}_{lu} = \mathcal{L}_l + \mathcal{L}_u \tag{15}$$

where $\mathcal{L}_l$ is the supervised loss for labeled data, and $\mathcal{L}_u$ is the unsupervised loss handling potential unknown categories. For labeled data, all pixels are of known categories, so the loss function is as follows:

$$\mathcal{L}_l = -\frac{1}{N}\sum_{i=1}^{N} y_i\log\left(\hat{y}_i\right). \tag{16}$$

For unlabeled data, since the image may contain unknown categories, the loss function is as follows:

$$\mathcal{L}_u = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i\log\left(\hat{y}_i\right) + (1 - y_i)\log\left(1 - \hat{y}_i\right)\right]. \tag{17}$$

N is the number of samples and y_i is the true label of the ith sample (0 or 1); $\hat{y}_i$ is the prediction probability for sample i.

The consistency loss $\mathcal{L}_{\text{con}}$ enforces stability in segmentation predictions by aligning the outputs of the teacher and student models on unlabeled data

$$\mathcal{L}_{\text{con}}(D_u, \theta_s) = -\frac{1}{|D_U||\Omega|}\sum_{x\in D_u}\sum_{\omega\in\Omega} c(\omega)\mathcal{L}\left(\widetilde{y}(\omega),\, p\theta_s(x)(\omega)\right). \tag{18}$$

Here, ω denotes the pixel address of the output lattice Ω of the segmentation map. $\widetilde{y}(\omega) \in 0, 1^Y$ and $p_{\theta_s}(x)(\omega) \in [0, 1]^Y$ represent the segmentation predictions from the teacher model and the student model, respectively. The term $c(\omega) \in [0, 1]$ indicates the segmentation prediction confidence from the teacher model at the pixel address ω, which is defined as

$$c(\omega) = \max\left(p_{\theta_t}(x)(\omega)\right). \tag{19}$$

This confidence score acts as a weighting factor to suppress unreliable pseudo-labels, ensuring that only high-confidence regions contribute significantly to the consistency loss.

IV. Experimental Results

A. Datasets

1) GID Dataset:

a) Labeled data: The GID-15 dataset [55] is a large-scale, high-resolution remote sensing land cover dataset derived from Gaofen-2 satellite data and released by Wuhan University. It comprises ten remote sensing images, each with dimensions of 7200 × 6800 pixels and a resolution of 2 m. The dataset is categorized into 15 land cover types, including industrial land (IDL), urban residential (UR) and rural residential (RR) areas, traffic land (TL), paddy fields (PFs), irrigated land (IL) and dry croplands (DCs), garden plots (GPs), arbor woodland (AW), shrub land (SL), natural grassland (NG), artificial grassland (AG), river (RV), lake (LK), and pond (PN) [4].

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TABLE I
CONVENTIONAL EXPERIMENTAL SETTINGS IN GID

Labeled ratio	1/2	1/4	1/8	1/16
Labeled number	1344	672	336	168
Unlabel number	1344	2016	2352	2520

TABLE II
NEW EXPERIMENTAL SETTINGS IN GID

Labeled ratio	1/2	1/4	1/8	1/16
Labeled number	1344	1344	1344	1344
Unlabel number	1344	4032	9408	201600

b) Unlabled data: The unlabeled data consists of publicly available datasets and additional Gaofen-2 data we collect. The public dataset refers to the GID-5 dataset, which includes 150 high-resolution Gaofen-2 images (7200 × 6800) obtained from over 60 cities in China, covering a geographical area of more than 50 000 km^2. Among these, ten scenes overlap with the GID-15 dataset, so only the remaining 140 scenes are used as unlabeled data in this study. Their access addresses are as follows: https://x-ytong.github.io/project/GID.html. In addition to the public dataset, we also collect Gaofen-2 data from the past five years, totaling 90 scenes, each with dimensions of 7200 × 6800. This data is publicly available at: https://pan.baidu.com/s/1ITqCQivYMRy8hQs9L1CyBQ. The unlabeled data used in the experiment are selected from this dataset.

c) Data settings: We crop each image into 210 non-overlapping image blocks with a size of 512 × 512. To verify the effectiveness of the proposed method, we divide the experimental settings into two types. The first aspect is a new experimental setting we suggest, where the amount of labeled data remains fixed while unlabeled data is continuously added (as shown in Table I). To ensure the diversity of the training set, we randomly select two or three image blocks from each labeled data as the training set (for a total of 24) and one image block from each scene as the test set (for a total of 10), with the remaining data designated as the validation set (a total of 6). The 24 images in the training set are utilized as labeled data for training. The corresponding unlabeled data is filtered from the unlabeled dataset. The second experimental setting is the conventional approach, where the total amount of data remains fixed while the ratio of labeled to unlabeled data is adjusted (as shown in Table II). The overall dataset includes 24 labeled image blocks along with the corresponding 24 unlabeled image blocks. The division between the test and validation sets remains the same as in the first setting.

2) LoveDA Dataset: The LoveDA dataset is an urban–rural domain adaptation surface cover dataset constructed by the RSIDEA team of the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing at Wuhan University. Its images were collected in Nanjing, Changzhou, and Wuhan and sourced from Google Earth. It contains 5987 high-resolution images with a resolution of 0.3 m, which are 8-bit deep and three-channel 1024 × 1024 images, and 166 768 labeled semantic objects. We crop each image into four non-overlapping image blocks with a size of 512 × 512. The label images are 8-bit deep and single-channel images. The categories include background, buildings, roads, water bodies, bare soil, forest land, and cultivated land, which are represented by pixel values from 1 to 7, respectively, and 0 represents the no-data area that should be ignored. This dataset has three characteristics: multiscale objects, complex background samples, and inconsistent class distributions. This data is publicly available at: https://github.com/Junjue-Wang/LoveDA.

TABLE III
NEW EXPERIMENTAL SETTINGS IN LOVEDA

Labeled ratio	1/2	1/4	1/8	1/16
Labeled number	462	462	462	462
Unlabel number	462	1386	3234	6930

This article is primarily dedicated to the research on the segmentation of urban areas. From the urban dataset, 115 images are filtered as labeled data. In addition, 338 images are selected to form the validation set, and another 339 images are used as the test set. The remaining data are uniformly regarded as the unlabeled dataset. All experiments are carried out in strict accordance with the new experimental settings presented in Table III. Each image is cut into four image patches of size 512 × 512 by means of sliding cropping and then input into the model for analysis.

B. Implementation Details

To ensure effective utilization of unlabeled data, the proposed framework first employs the CMM to assess the distribution similarity between unlabeled and labeled data, ranking the unlabeled samples accordingly. Based on this ranking, a subset of unlabeled data is selected and, together with labeled data, fed into the CB4S network, where the proposed training strategy is applied to enhance segmentation performance while mitigating semantic drift. We use three metrics to evaluate our method's performance: mean intersection over union (mIOU), $F1$-score, and overall accuracy (OA).

The batch size is 16, with eight samples from labeled data and eight from unlabeled data. The teacher and student models are pre-trained on labeled data for five epochs before training begins. The clustering learning method uses a stochastic gradient descent optimizer with the following settings: a base learning rate of 0.01, a momentum of 0.9, and a weight decay of 0.0001. Training spans 40 epochs, starting with an initial learning rate of 0.01. After each iteration, the learning rate is updated by a factor of $(1 - (\text{iter}/\text{total}_{\text{iter}}))^{0.9}$. The momentum coefficient is set to 0.9, and the weight decay coefficient is set to 0.0001. The network was implemented in PyTorch and trained on a TITAN RTX GPU with 24 GB of memory. During training, both the teacher and student models

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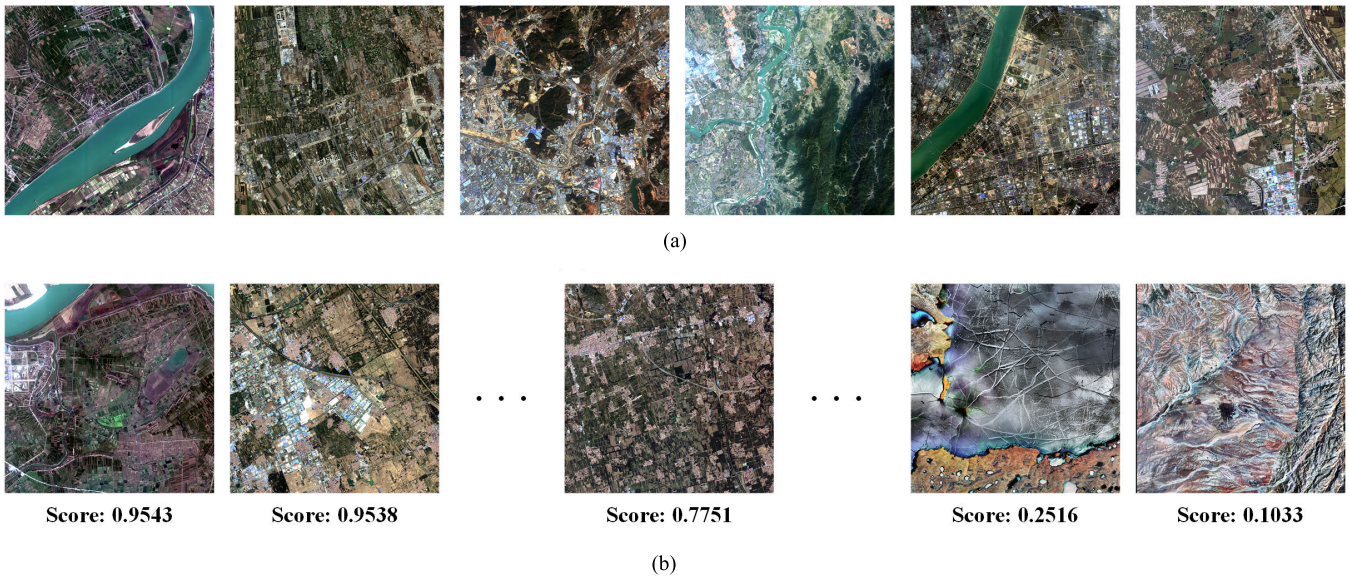


Fig. 4. Sorting results of unlabeled data. (a) Labeled data. (b) Ranked unlabeled data.

TABLE IV
EFFECT OF UNLABELED DATA QUALITY ON MODEL PERFORMANCE IN GID DATASET UNDER CONVENTIONAL EXPERIMENTAL SETTING

Total data	Label data (label ratio)	UnLabel data	Model Name	Evalution index		
				OA	mIOU	Avg.F1
2688	1344 (1/2)	1344	Worst	72.24	43.58	57.2
			Random	72.93	43.99	57.28
			Best	**73.1**	**44.15**	**57.83**
	672 (1/4)	2016	Worst	70.53	40.12	52.34
			Random	71.69	42.88	55.47
			Best	**72.03**	**44.45**	**57.91**
	336(1/8)	2352	Worst	62.43	25.93	35.68
			Random	63.83	26.86	37.00
			Best	**64.97**	**27.84**	**37.86**
	168 (1/16)	2520	Worst	50.72	17.68	25.93
			Random	55.6	17.03	24.57
			Best	**59.34**	**20.91**	**29.44**

TABLE V
EFFECT OF UNLABELED DATA QUALITY ON MODEL PERFORMANCE IN GID DATASET UNDER NEW EXPERIMENTAL SETTING

Total data	Label data (label ratio)	UnLabel data	Model Name	Evalution index		
				OA	mIOU	Avg.F1
2688	1344 (1/2)	1344	Worst	72.24	43.58	57.2
			Random	72.93	43.99	57.28
			Best	**73.1**	**44.15**	**57.83**
5376	1344 (1/4)	4032	Worst	72.77	44.58	57.56
			Random	73.68	45.58	58.66
			Best	**74.29**	**47.12**	**60.19**
10752	1344 (1/8)	9408	Worst	73.45	45.01	58.17
			Random	73.75	44.32	57.15
			Best	**74.54**	**45.98**	**58.39**
21504	1344 (1/16)	201600	Worst	73.03	45.76	57.57
			Random	73.4	47.01	59.28
			Best	**74.06**	**47.43**	**60.15**

are jointly trained, while only the teacher model is used for prediction during testing and validation.

C. Ablation Study

To evaluate the proposed method's performance, we use the Mean Teacher as the baseline in this study and conduct experimental validation under both conventional and novel settings. Since the optimization strategy is primarily designed for large-scale remote sensing datasets, the final ablation experiments are conducted exclusively under the new experimental setting. Besides, based on empirical observations, the $\text{initial}_{\text{amount}}$ of unlabeled data is set to three times the labeled data, as this proportion maintains stable model performance and serves as a reliable baseline for progressive training.

1) Effects of the Unlabeled Data Quality: The performance of semi-supervised semantic segmentation heavily depends on the quality of unlabeled data. Using existing labeled data, we employ the CCM module to sort all unlabeled data, with the sorting results displayed in Fig. 4. As shown in the figure, despite being captured by the same satellite, the data exhibit substantial quality variations due to geographical location, imaging time, and weather conditions. By applying the proposed method, images that exhibit substantial distributional differences from the labeled data can be effectively filtered out. Specifically, following the computation method described in Section III-B, the scores of different unlabeled images are obtained and ranked in descending order. The ranking results align well with the visual intuition of humans, further validating the effectiveness of the proposed method.

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TABLE VI

EFFECT OF UNLABELED DATA QUALITY ON MODEL PERFORMANCE IN LOVEDA DATASET UNDER NEW EXPERIMENTAL SETTING

Total data	Label data (label ratio)	UnLabel data	Model Name	Evalution index		
				OA	mIOU	Avg.F1
932	462 (1/2)	462	Worst	35.47	17.72	27.33
			Random	47.14	31.63	44.43
			Best	**59.08**	**40.02**	**54.83**
1848	462 (1/4)	1386	Worst	44.47	29.52	41.77
			Random	51.76	35.98	50.53
			Best	**60.08**	**42.35**	**56.89**
3696	462 (1/8)	3234	Worst	42.17	26.02	38.13
			Random	54.75	39.08	52.54
			Best	**59.97**	**42.79**	**57.61**
7392	462 (1/16)	6930	Worst	55.1	37.74	51.72
			Random	55.14	39.16	53.78
			Best	**59.91**	**42.38**	**56.12**

TABLE VII

GID: EFFECT OF CONSISTENCY-BASED SEMI-SUPERVISED SEMANTIC SEGMENTATION MODEL IN CONVENTIONAL EXPERIMENTAL SETTINGS

Total data	Label data (label ratio)	UnLabel data	Model Name	Evalution index		
				OA	mIOU	Avg.F1
2688	1344 (1/2)	1344	Baseline	73.10	44.15	57.83
			Ours	**73.94**	**44.94**	**58.68**
	672 (1/4)	2016	Baseline	72.03	44.45	57.91
			Ours	**72.76**	**44.97**	**58.50**
	336 (1/8)	2352	Baseline	64.97	27.84	37.86
			Ours	**65.5**	**29.81**	**40.56**
	168 (1/16)	2520	Baseline	58.97	20.06	28.34
			Ours	**59.34**	**20.91**	**29.44**

To systematically assess the impact of the quality of unlabeled data on segmentation performance, we select the data with the highest (best) and lowest (worst) score and randomly sampled data (random), and perform experimental validation in both conventional and our novel experimental settings. The evaluation is conducted on the GID and LoveDA datasets, with the results presented in Tables IV and V for GID, and Table VI for LoveDA. The experimental results demonstrate that under both the newly proposed and traditional experimental settings, using the highest-scoring unlabeled data consistently yields the best performance. Selecting the data with the highest score consistently yields the best results in both settings. In particular, for the conventional setting, comparing the best and worst scores across 1/2, 1/4, 1/8, and 1/16 reveals performance improvements of 0.77%, 1.50%, 2.54%, and 8.62%, respectively. As labeled data becomes scarcer, the quality of unlabeled data has an increasingly significant impact on model performance. These results highlight the influence of unlabeled data quality on semantic segmentation performance and validate the effectiveness of our data screening method.

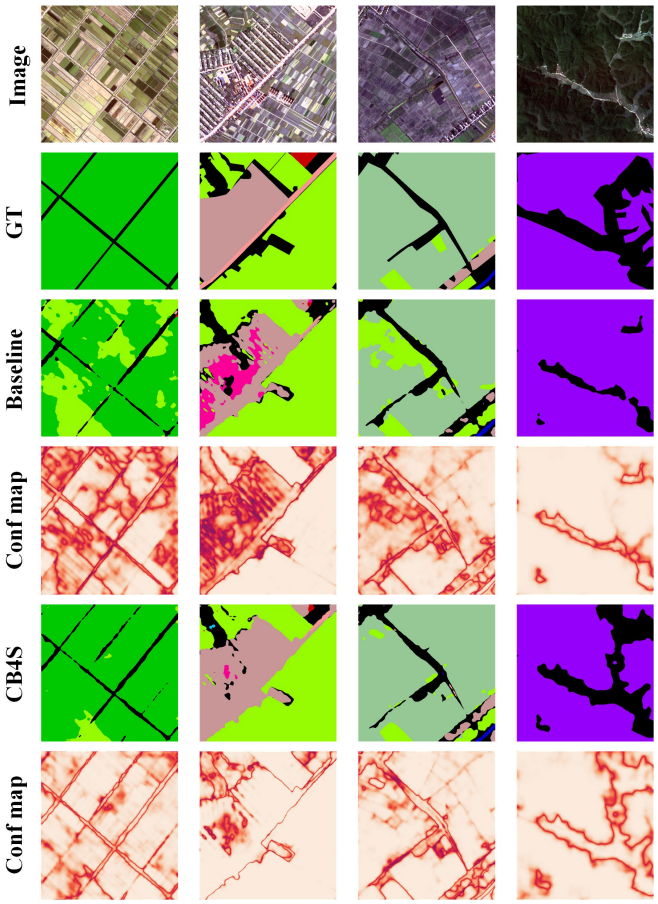


Fig. 5. Prediction results and their corresponding confidence map for the consistency-based semi-supervised semantic segmentation framework. Among them, CB4S stands for consistency-based semi-supervised semantic segmentation network, and conf stands for confidence map.

TABLE VIII

GID: EFFECT OF CONSISTENCY-BASED SEMI-SUPERVISED SEMANTIC SEGMENTATION MODEL IN NEW EXPERIMENTAL SETTINGS

Total data	Label data (label ratio)	UnLabel data	Model Name	Evalution index		
				OA	mIOU	Avg.F1
2688	1344 (1/2)	1344	Baseline	73.10	44.15	57.83
			Ours	**73.94**	**44.94**	**58.68**
5376	1344 (1/4)	4032	Baseline	74.29	47.12	60.19
			Ours	**74.94**	**47.67**	**60.35**
10752	1344 (1/8)	9408	Baseline	74.54	45.98	58.39
			Ours	**74.80**	**47.19**	**59.68**
21504	1344 (1/16)	201600	Baseline	74.06	47.41	60.01
			Ours	**74.3**	**47.43**	**60.15**

It is worth noting that the effectiveness of our method relies on the assumption that the distribution shift between labeled and unlabeled data is not excessively large. When the distribution discrepancy is too significant (e.g., extremely high MMD scores), even after filtering, the selected unlabeled data may still introduce noise into the model training. This limitation suggests that our approach is more suitable for cases where the labeled and unlabeled data share some underlying similarity,

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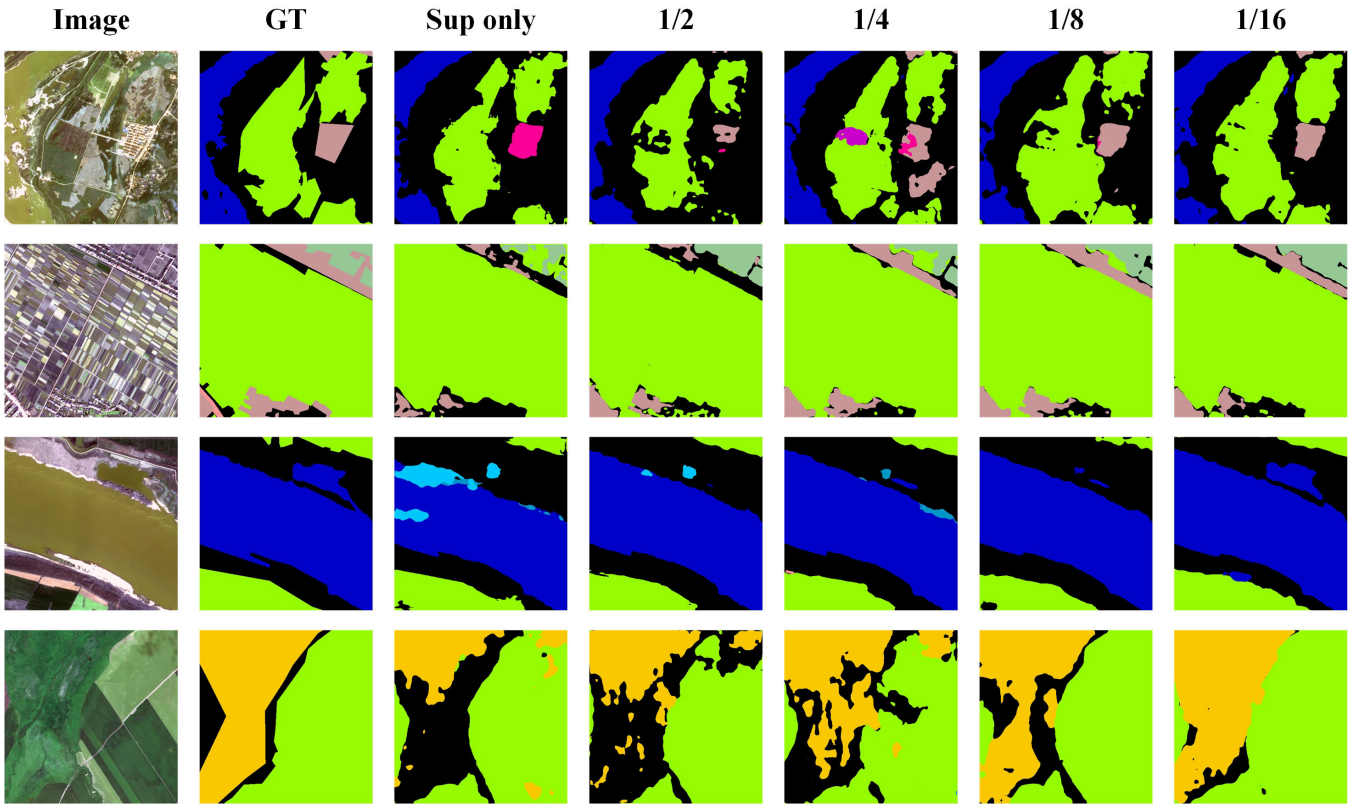


Fig. 6. Prediction results with different amounts of training data based on our new experimental settings.

TABLE IX

LoveDA: Effect of Consistency-Based Semi-Supervised Semantic Segmentation Model in New Experimental Settings

Total data	Label data (label ratio)	UnLabel data	Model Name	Evalution index		
				OA	mIOU	Avg.F1
924	462 (1/2)	462	Baseline	59.08	40.02	54.83
			Ours	**59.29**	**42.39**	**56.85**
1848	462 (1/4)	1386	Baseline	60.08	42.35	56.89
			Ours	**60.88**	**42.83**	**57.37**
3696	462 (1/8)	3234	Baseline	59.97	42.79	57.61
			Ours	**60.62**	**43.16**	**58.58**
7392	462 (1/16)	6930	Baseline	59.91	42.38	56.12
			Ours	**61.53**	**43.77**	**57.79**

TABLE X

Effect of Model Training Optimization Strategies in New Experimental Settings

Label data (label ratio)	Unlabel	Optimization strategy	Evalution index		
			OA	mIOU	Avg.F1
Sup only		N	71.53	42.04	56.06
462 (1/2)	462	N	73.94	44.94	58.68
462 (1/4)	1386	N	74.94	47.67	60.35
462 (1/8)	3234	N	74.8	47.19	59.68
		Y	74.99	48.44	61.57
462 (1/16)	6930	N	74.3	47.43	60.15
		Y	**75.27**	**49.45**	**62.61**

such as being acquired from similar geographical regions or sensor types. Future work could explore adaptive strategies to handle scenarios with more severe distribution shifts, such as combining MMD with feature clustering techniques.

2) Effects of Consistency-Based Semi-Supervised Semantic Segmentation Model: We design a consistency-based semi-supervised semantic segmentation network to further mitigate the interference of unknown categories caused by unlabeled data. The experimental results under both the conventional and new experimental settings are presented in Tables VII and VIII, respectively. In both settings, all data are selected from images with the highest data screening scores, and the data used in both settings are identical when the label ratio is 1/2. First, we examine the results under the conventional experimental settings. As the labeled-to-unlabeled data ratio is adjusted, the model's OA improves by 0.84%, 0.73%, 0.53%, and 0.37%, respectively. Next, we analyze the results from the experimental settings proposed in this article. With a fixed amount of labeled data, as the quantity of unlabeled data increases, the OA improves by 0.84%, 0.65%, 0.26%, and 0.24%, respectively. From these two sets of experiments, we observe that, regardless of whether the conventional or our proposed experimental setting is used, the method presented in this article effectively alleviates the impact of unknown categories. As shown in Table IX, the results on the LoveDA

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TABLE XI
LOVEDA: EFFECT OF MODEL TRAINING OPTIMIZATION STRATEGIES IN NEW EXPERIMENTAL SETTINGS

Label data (label ratio)	Unlabel	Optimization strategy	Evalution index		
			OA	mIOU	Avg.F1
Sup only		N	51.99	35.16	50.7
462 (1/2)	462	N	59.29	42.39	56.85
462 (1/4)	1386	N	60.88	42.83	57.37
462 (1/8)	3234	N	60.62	43.16	58.58
		Y	62.19	44.42	58.59
462 (1/16)	6930	N	61.53	43.77	57.79
		Y	**62.88**	**45.09**	**59.33**

TABLE XII
RESULT OF OUR METHOD ON RS_DWL

total	Label data (label ratio)	UnLabel data	Model	Evalution index		
				OA	mIOU	Avg.F1
924	462 (1/2)	462	Random	50.03	36.52	50.35
			Best	51.33	39.48	53.31
			+info	**55.44**	**39.64**	**53.89**
460	462 (1/4)	1386	Random	61.53	44.54	60.21
			Best	62.45	46.47	62.42
			+info	**63.17**	**47.02**	**62.33**
920	462 (1/8)	3234	Random	61.95	46.61	61.98
			Best	63.57	46.92	62.4
			+info	63.86	47.1	62.69
			+info+str	**64.01**	**47.45**	**62.93**
1840	462 (1/16)	6930	Random	58.6	44.3	59.84
			Best	62.13	46.08	61.59
			+info	62.63	46.45	62.08
			+info+str	**64.31**	**48.95**	**64.98**

TABLE XIII
RESULT OF OUR METHOD ON WSCL

total	Label data (label ratio)	UnLabel data	Model	Evalution index		
				OA	mIOU	Avg.F1
924	462 (1/2)	462	Random	60.56	44.2	59.94
			Best	62.54	46.26	62.13
			+info	62.88	46.64	62.39
460	462 (1/4)	1386	Random	61.96	44.87	60.45
			Best	62.76	46.13	61.68
			+info	63.34	46.98	62.17
920	462 (1/8)	3234	Random	62.3	45.85	61.18
			Best	62.81	46.02	61.52
			+info	63.49	46.12	61.78
			+info+str	64.21	47.53	63.00
1840	462 (1/16)	6930	Random	61.24	44.83	59.57
			Best	63.74	46.28	60.79
			+info	64.42	47.36	62.87
			+info+str	64.79	47.84	63.16

TABLE XIV
PARAMETERS OF DIFFERENT MODELS

img_size	ModelName	Evalution index		
		Parameters/M	Flops/GMac	FPS
512*512	Meanteacher	83.52	158.66	10.51
	Meanteacher_our	84.81	179.86	9.15
	RS_DWL	41.76	79.33	6.04
	RS_DWL_our	43.05	100.53	5.43
	WSCL	42.49	82.54	2.97
	WSCL_our	43.78	103.74	2.42

dataset indicate that, compared to the Baseline, mitigating the interference of unknown categories leads to a progressive improvement in overall model accuracy as the amount of unlabeled data increases. Specifically, the accuracy gains are 0.21%, 0.8%, 0.65%, and 1.62%, respectively. The proposed method consistently achieves the best performance across different datasets and experimental settings.

To provide a more intuitive demonstration of the effectiveness of the proposed model, we compare its prediction results with those of the baseline model, along with their corresponding confidence maps, as shown in Fig. 5. The figure illustrates that after mitigating the interference of unknown categories, the proposed model achieves more precise decision boundaries and effectively distinguishes between easily confused categories. Additionally, the results indicate that as the proportion of unlabeled data increases, the performance improvement of the model becomes less significant. Moreover, as observed in the tables, the model achieves optimal performance when the amount of unlabeled data is three times that of the labeled data. However, when the ratio increases to seven and fifteen times, the model's performance begins to decline noticeably. This finding suggests that as the volume of unlabeled data increases, the issue of semantic drift remains a critical factor affecting segmentation performance.

3) Effects of Optimization of Training Strategy: As discussed above, in large-scale and complex scenarios, as the amount of unlabeled data increases, model's performance is impacted by semantic drift, leading to a decline in performance. This article employs optimization strategies to mitigate the semantic drift problem in large-scale datasets. Therefore, experimental verification is conducted using only the experimental settings proposed in this article, specifically when the amount of unlabeled data is seven and 15 times that of the labeled data. The results are shown in Tables X and XI, where "*N*" indicates the absence of the optimization training strategy, and "*Y*" indicates that it is used. As shown in the tables, with the optimized training strategy, the model achieves accuracy improvements of 0.19% and 0.97% on the GID dataset and 1.57% and 1.35% on the LoveDA dataset when the amount of unlabeled data is seven and 15 times that of the labeled

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TABLE XV
GID: COMPARATIVE EXPERIMENT WITH DIFFERENT UNLABELED DATA AMOUNT IN NEW EXPERIMENT SETTINGS

total	Label (ratio)	UnLabel data	Model	Evalution index		
				OA	mF1	mIou
1344	super only		Baseline	71.53	42.04	56.06
2688	1344 (1/2)	1344	Baseline	72.93	43.99	58.68
			CPS	73.37	44.67	57.57
			GCT	73.52	44.15	58.61
			CCT	71.54	44.45	-
			PS-MT	73.54	44.17	58.68
			fixmatch	71.24	42.51	56.97
			RS_DWL	73.33	44.24	59.06
			BCF	73.25	44.01	58.88
			Ours	**73.94**	**44.94**	**59.34**
5376	1344 (1/4)	4032	Baseline	73.68	45.58	58.66
			CPS	74.16	46.08	58.58
			GCT	74.13	46.27	59.68
			CCT	72.5	43.41	-
			PS-MT	74.38	46.77	59.31
			fixmatch	73.93	46.35	58.95
			RS_DWL	74.66	47.42	59.88
			BCF	74.57	47.29	59.53
			Ours	**74.94**	**47.67**	**60.35**
10752	1344 (1/8)	9408	Baseline	73.75	44.32	57.15
			CPS	73.23	46.29	58.67
			GCT	73.67	46.68	59.57
			CCT	71.9	43.25	-
			PS-MT	72.85	45.98	58.32
			fixmatch	72.32	44.85	57.81
			RS_DWL	73.51	46.21	59.76
			BCF	73.84	47.55	60.07
			Ours	**74.99**	**48.44**	**61.57**
21504	1344 (1/16)	201600	Baseline	73.40	47.01	59.28
			CPS	71.92	44.09	56.77
			GCT	73.33	46.38	58.94
			CCT	72.68	43.22	-
			PS-MT	73.17	45.62	58.22
			fixmatch	72.31	44.51	57.53
			RS_DWL	73.44	47.23	59.64
			BCF	73.27	46.99	60.25
			Ours	**75.27**	**49.45**	**62.61**

TABLE XVI
LOVEDA: COMPARATIVE EXPERIMENT WITH DIFFERENT UNLABELED DATA AMOUNT IN NEW EXPERIMENT SETTINGS

total	Label (ratio)	UnLabel data	Model	Evalution index		
				OA	mF1	mIou
1344	super only		Meanteacher	51.99	35.16	50.7
924	462 (1/2)	462	Meanteacher	35.47	17.72	27.33
			Fixmatch	56.27	39.68	56.2
			Unimatch	60.96	43.91	59.92
			RS_DWL	50.03	36.52	50.35
			BCF	58.84	43.22	59.15
			WSCL	60.56	44.2	59.94
			Meanteacher_our	59.29	42.39	56.85
			RS_DWL_our	55.44	39.64	53.89
			WSCL_our	**62.88**	**46.64**	**62.39**
1848	1344 (1/4)	1386	Meanteacher	44.47	29.52	41.77
			Fixmatch	52.49	34.86	50.03
			Unimatch	60.73	45.40	61.68
			RS_DWL	61.53	44.54	60.21
			BCF	60.60	44.35	59.94
			WSCL	61.96	44.87	60.45
			Meanteacher_our	60.88	42.83	57.37
			RS_DWL_our	63.17	47.02	62.33
			WSCL_our	**63.34**	**46.98**	**62.17**
3696	462 (1/8)	3234	Meanteacher	42.17	26.02	38.13
			Fixmatch	57.91	41.64	57.75
			Unimatch	62.54	46.13	62.29
			RS_DWL	61.95	47.1	62.69
			BCF	61.42	46.55	62.06
			WSCL	62.3	45.85	61.18
			Meanteacher_our	62.19	44.42	58.59
			RS_DWL_our	64.01	47.45	62.93
			WSCL_our	**64.21**	**47.53**	**63**
7392	462 (1/16)	6930	Meanteacher	55.1	37.74	51.72
			Fixmatch	57.66	37.4	52.03
			Unimatch	62.26	44.87	59.44
			RS_DWL	58.6	44.3	59.84
			BCF	58.62	43.83	59.61
			WSCL	61.24	44.83	59.57
			Meanteacher_our	62.88	45.09	59.33
			RS_DWL_our	64.31	48.95	64.98
			WSCL_our	**64.79**	**47.84**	**63.16**

data, respectively. This indicates that the proposed strategy, by gradually introducing more challenging data, prevents the model from overfitting low-quality pseudo-labels in the early training stages, thereby enhancing its robustness and generalization ability.

Furthermore, the effectiveness of the optimization strategy is further validated through the visualization of prediction results, as illustrated in Fig. 6. The figure demonstrates that as the amount of unlabeled data increases, the model trained with the optimization strategy exhibits progressively improved performance in complex scenes, particularly in terms of decision boundary clarity and fine-detail preservation. This observation suggests that the optimization strategy enhances the model's adaptability to complex unlabeled data, enabling it to better learn and integrate diverse feature distributions, ultimately reducing the negative impact of semantic drift on segmentation performance.

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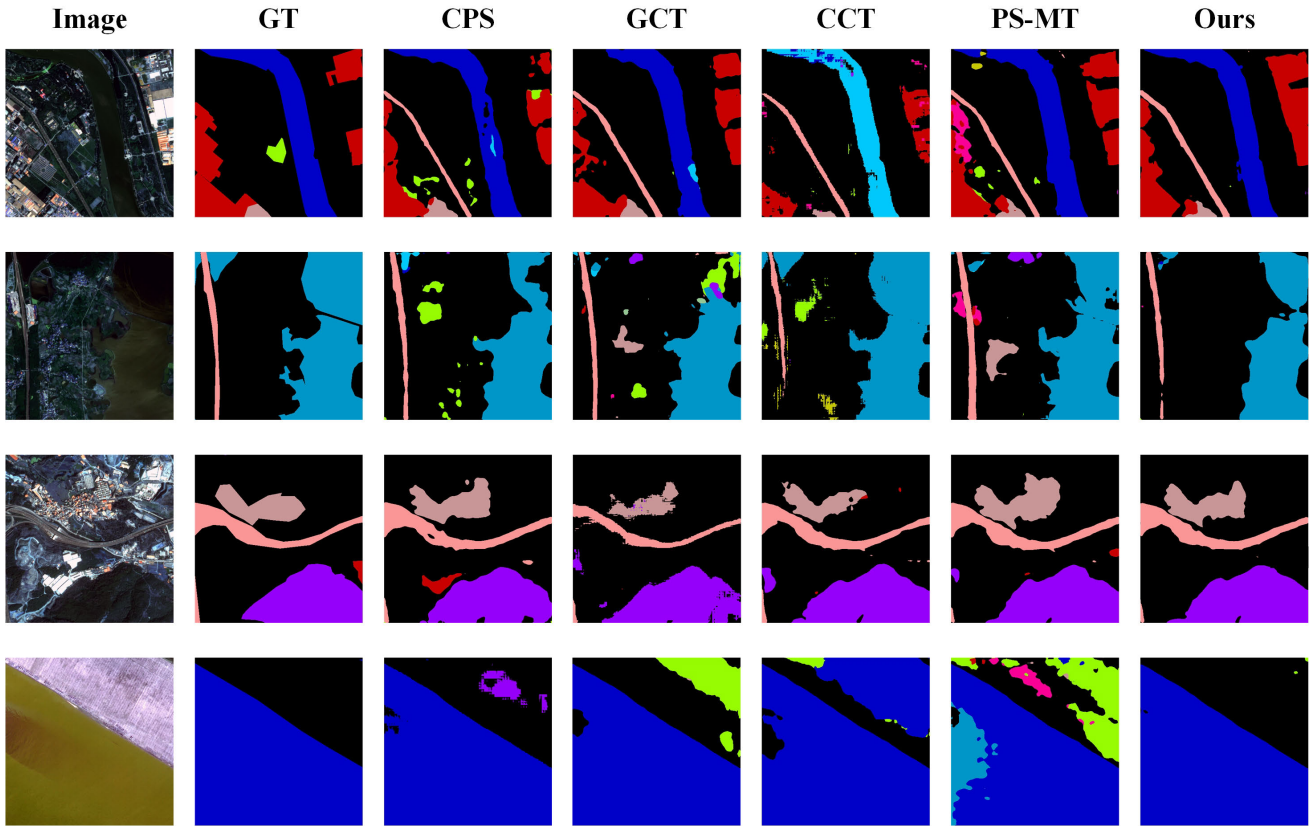


Fig. 7. Prediction results and their corresponding confidence map for the consistency-based semi-supervised semantic segmentation framework.

4) Effects of the Proposed Method Across Different Baselines: To further validate the effectiveness of the proposed method, experiments were conducted on the LoveDA dataset using different baseline models. The results are presented in Tables XII and XIII. By comparing RS_DWL_Random [13] and WSCL_Random [56] under different amounts of unlabeled data, it can be observed that when using randomly selected unlabeled data, both models exhibit a performance degradation trend. This further confirms the challenge of maintaining robust performance in remote sensing semantic segmentation for complex scenes. Furthermore, when incorporating the proposed method into these baselines, the results indicate that our approach remains effective across different baseline models. Comparing $*$_Random and $*$_Best results further demonstrates that the quality of unlabeled data significantly impacts different models. The selection strategy proposed in this study effectively enhances model performance by ensuring the use of high-quality unlabeled data.

5) Parameters of Different Models: To evaluate the parameter count and training speed of different models, we select Mean Teacher, RS_DWL, and WSCL as representative examples. Both the original models and their counterparts incorporating the proposed method ($*$_our) are assessed in terms of model parameters, FLOPs (floating point operations per second), and FPS (frames per second). As shown in Table XIV, introducing the proposed method results in a moderate increase in model parameters and FLOPs, while FPS exhibits a slight decrease. This indicates that although the proposed method enhances the semantic segmentation performance, it also increases model complexity and computational demands, potentially leading to longer training and inference times. Therefore, when designing and optimizing the model, a trade-off between performance improvement and computational cost must be carefully considered.

D. Comparison With Previous Methods

To demonstrate the superiority of the proposed method, this section compares it with several state-of-the-art semi-supervised semantic segmentation methods, including CPS [29], GCT [57], CCT [40], PS-MT [58], FixMatch, UniMatch, BCF [59], RS_DWL, and WSCL. Given that the proposed approach is primarily designed for large-scale real-world scenarios, the comparisons are conducted under a fixed amount of labeled data while varying the amount of unlabeled data. The detailed results are presented in Tables XV and XVI.

1) Comparative Experiments on the GID Dataset: By comparing different amounts of unlabeled data, our method achieves the best results in OA, mIoU, and Avg.$F1$. To more intuitively demonstrate the effectiveness of our process, we visualize the prediction results of different models using 1/16 of the data as an example, as shown in Fig. 7. The figure shows that our full-cycle framework effectively distinguishes the correct category in complex regions. Our method maintains strong segmentation performance, particularly when confronted with interfering categories, demonstrating its robustness. Furthermore, by comparing the growth in model performance under varying amounts of data, we observe that our model's performance becomes more stable relative to other

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models as the amount of data increases. This performance improvement stems from three key components in our framework. First, the CMM helps mitigate noise from inconsistent data, reducing the impact of unknown category interference and semantic drift. Second, the consistency-based segmentation network improves the model's ability to handle unknown categories by isolating and separating unknown regions in the feature space. Third, the progressive batch training strategy gradually introduces more complex unlabeled data during training, effectively mitigating semantic drift. Further analysis shows that our method consistently outperforms other models as the amount of unlabeled data grows. For instance, when the amount of unlabeled data matches the labeled data, our model shows an improvement of at least 0.4% over other models. When the amount of unlabeled data is three times that of the labeled data, the improvement increases to 0.56%, and when it is seven times the labeled data, the improvement is 1.24%. When the amount of unlabeled data reaches 15 times that of the labeled data, our model improves by at least 1.87%. These results further confirm that our method maintains stable performance even when faced with large-scale, complex remote sensing scenes without degrading as the amount of unlabeled data increases. In conclusion, the method presented in this article effectively mitigates the risk of positional category interference and semantic drift in complex unlabeled data while ensuring model stability. It enhances the model's ability to adapt to potential data changes, improving overall generalization performance.

2) Comparative Experiments on the LoveDA Dataset: Table XVI presents the segmentation results of various methods on the LoveDA dataset, where $*$_our denotes the results obtained by applying the proposed method to the respective models. As shown in the table, the proposed method consistently achieves the best performance in terms of OA, mIoU, and Avg.$F1$ as the amount of unlabeled data increases. Furthermore, the results indicate that models incorporating the proposed approach exhibit greater performance improvements with increasing unlabeled data, whereas other methods perform significantly worse. This further validates the effectiveness of the proposed method. The overall experimental results demonstrate that the proposed method effectively mitigates performance degradation in the semantic segmentation of remote sensing images under complex scenarios. Compared with conventional semi-supervised semantic segmentation methods, the proposed approach places greater emphasis on the quality of remote sensing data, accounts for the inherent differences between labeled and unlabeled data, and incorporates full-cycle optimization from both model design and training strategy perspectives. The results further confirm that the proposed method consistently improves segmentation performance and stability across different remote sensing datasets, highlighting its effectiveness and robustness in semi-supervised remote sensing semantic segmentation tasks.

V. Conclusion

This article investigates the key factors contributing to performance degradation as the amount of unlabeled data increases, focusing on three critical aspects: data, model, and training strategy. To address these challenges, we propose a comprehensive full-cycle semi-supervised semantic segmentation framework. The framework includes an unlabeled data screening mechanism that ensures model stability at the data level, a consistency-based approach to mitigate interference from unknown categories in remote sensing imagery at the model level, and an optimized training strategy to reduce the impact of semantic drift. By tackling these challenges across multiple levels, our framework ensures the stability, robustness, and overall performance of the semi-supervised semantic segmentation model. Furthermore, we introduce a new experimental setup and conduct extensive ablation experiments to evaluate the effectiveness of each component within our framework. The experimental results on the widely used GID-15 and LoveDA datasets demonstrate that the proposed framework outperforms traditional methods, highlighting its potential for handling complex remote sensing tasks.

In conclusion, this work offers a comprehensive approach to tackling key challenges in semi-supervised semantic segmentation, laying a foundation for future research in handling larger datasets and more complex real-world scenarios. Potential future directions include improving computational efficiency, adapting to diverse geographic and temporal data, and expanding the framework to handle even larger-scale and more heterogeneous datasets.

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